9 November 2026

Editor-in-Chief

Journal of Seismic Exploration

Dear Editor,

Re: Submission of "Higher-mode Love-wave tracking and reliability for coal-seam thickness inversion"

I am pleased to submit the enclosed original research article for consideration by the Journal of Seismic Exploration.

The study evaluates how much information higher-mode Love waves add for coal-seam thickness inversion, how much of that information survives automatic dispersion tracking, and whether the resulting parameter intervals are credible. Its main contribution is an integrated information-retention and reliability analysis rather than another multi-frequency thickness workflow. The experiments use fixed physical mode identities, noisy dispersion images, robust likelihoods, split-conformal calibration, independent finite-difference wavefields, a QEDispInv root comparison and a standard-linear-solid attenuation check.

The manuscript reports both positive and negative findings. Higher modes substantially improve theoretical precision, but their benefit is limited by mode availability, automatic-pick reliability and cross-domain calibration. The title, abstract and conclusions explicitly identify the work as a synthetic benchmark and do not claim field accuracy.

The author confirms that the manuscript is original, has not been published previously, and is not under consideration by another journal. Lu Bin is the sole author and approves this submission. The author declares no competing interests. This work was supported by the National Natural Science Foundation of China (Grant No. 41974209).

The code and synthetic results are publicly available at https://github.com/Lubin2000/coal-seam-love-wave-reliability. The version-family concept DOI is https://doi.org/10.5281/zenodo.24054133. The author used OpenAI Codex for coding support, literature organization, numerical experiment assistance and language editing; all content was reviewed and verified, and the complete declaration is included in the back matter.

Thank you for your time and consideration.

Sincerely,

Lu Bin

Wenzhou University of Technology

lubin2000@163.com